Testing the reproducibility and robustness of "Cognitive causes of 'like me' race and gender biases in human language production"

Marie Klünter, Rose Hörsting, Luca Leppert, Gina Reinhard & Elen Le Foll

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Abstract

The original study by Brough et al. (2024) investigates how social biases may influence automatic word retrieval mechanisms during sentence production. Across several experimental studies, the authors observe significant 'like me' effects, whereby speakers preferentially mention people who share their own race and gender first and faster than out-group members. Psycholinguistic manipulations show that the effect does not vary across grammatical constructions. We conducted a computational reproduction using the authors' original raw data and R scripts and performed multiple robustness checks (alternative model specifications, alternative exclusion criteria, etc.). We conclude that the findings reported in Brough et al. (2024) are computationally reproducible using the authors' original scripts with some minor adjustments and additions. We also found them to be robust to our modifications. However, the reported effects supporting the paper's main claims are very small, which the article does not make sufficiently clear and no correction for multiple testing appears to have been applied. Moreover, the reported analyses for Studies 2 and 3 differ considerably from the preregistered ones and the key multi-study analysis does not appear to have been preregistered. To facilitate future repetitive research and cumulative endeavours, we formulate a number of recommendations including clearly distinguishing confirmatory from exploratory analyses, transparently reporting deviations from preregistered analysis plans, and consistently reporting effect sizes in interpretable units.

Keywords: language production, psycholinguistics, race bias, gender bias, lexical retrieval

1 Background

In "Cognitive causes of 'like me' race and gender biases in human language production", Brough et al. (2024) describe how speakers are more likely to mention individuals who resemble their social group first. They demonstrate that the names of individuals who are more similar to the participants are more accessible to the participants and conclude that this 'like me' effect is due to "(…) automatized mechanisms by which humans retrieve words from memory to produce sentences" (Brough et al. 2024: 1706).

The original article examines the influence of both gender and race across four independent studies. In Studies 1–3, participants had to learn the names of eight persons shown on photographs, who were either Black or white, and male or female. They then had to describe events in which two persons, differing in either gender or race, engaged in symmetrical interactions. The studies aimed to measure "the degree to which participants framed their descriptions in ways that highlighted their own social group, by beginning their sentences with the name of the person who matched them in gender or in race" (Brough et al. 2024: 1707). Based on the results from experiments testing both English L1 and L2 speakers as well as Chinese L1 speakers, Brough et al. (2024: 1710) claim that this social bias "not only holds consistently across different linguistic constructions and different languages, but that [it] also holds similarly across different social groups".

Although the overall 'like me' effect was small, with mean values of around 55% (Brough et al. 2024: 1711), the authors conclude that the observed social bias is important. They suggest that the effect is independent of language, linguistic construction, and social group, although Brough et al. (2024: 1711) acknowledge that the first conclusion is based on just two languages (English and Chinese). According to the authors, "the fact that the like me effect was invariant across constructions suggests that it is driven by accessibility and incrementality, rather than by evaluation" (Brough et al. 2024: 1711). They claim that "a core conclusion from the present studies is that the incremental and accessibility-driven language system that allows speakers to produce language that is fluent also leads them to produce language that is biased" (Brough et al. 2024: 1711).

1.1 Motivation for the present study

Although recently published, the original study can be said to have been influential as it has already been cited in an article on 'Revisiting the policy implications of implicit social cognition' (Axt et al. 2025), which concludes with a hope that more studies will examine the large phenomenon of implicit biases in the future. Moreover, it has been cited in recent AI-related publications, e.g., on how interactions with counter-stereotypical AI systems can be used to shift our stereotypical representations (Hermann, De Freitas, and Puntoni 2025) and on social perceptions of faces in vision-language models (Hausladen et al. 2025).

In a recent Bachelor thesis, Halpern (2026) attempted to replicate the 'like me' bias finding using animated characters in addition to still image stimuli as in the original study. The study failed to replicate a significant overall 'like me' effect, but reported a significant gender effect. Male participants were more likely to produce a 'like me' bias than female participants. Moreover, men were more likely to produce a 'like me' bias in the animated condition, whereas there was no difference between women and men in the mixed condition. A major limitation of this replication is its small and WEIRD (Andringa and Godfroid 2020; Henrich, Heine, and Norenzayan 2010) sample, which consisted of only 32 psychology students from Florida State University (Halpern 2026).

To the best of our knowledge, the five studies presented in Brough et al. (2024) have not yet been reproduced although this is feasible because the authors uploaded their data, stimuli, and R scripts on the OSF under https://osf.io/87vew/ (28 November 2028). In addition, we were able to consult the preregistrations, as well as the peer review reports, which the journal, *Nature Human Behaviour*, published alongside the article. We began working on the computational reproduction part of our analyses as part of the Replications Games organised by the Institute for Replication at the University of Münster on 28 November 2025.

1.2 Reproduced Study, Hypothesis, and Findings

We set out to computationally reproduce all the descriptive and inferential statistics, statistical models, and plots reported in the original publication.

Our robustness checks test the robustness of the findings used to support hypothesis….

2 Methods

The present study started as an exact computational reproduction, which we conducted in R, the programming language also used by the original authors, based on the authors' original raw data and - as far as possible (see Section 3) - original preprocessing and analysis scripts.

Examining the preregistrations of the original studies (see Section 3.1), we noticed that the reported models in Studies 2 and 3 differ considerably from the models that were preregistered. We therefore decided to perform robustness checks by fitting the preregistered models to the original data provided by the authors. In addition, we decided to test the claim made in the discussion of the results of Study 4 that the observed 'like me' gender effects may be smaller in Chinese due to the inclusion of two potentially gender-neutral Chinese names in the experiment (Brough et al. 2024: 1713) by rerunning the authors' analysis with a subset of the data that excludes these two names. We compare the results of the reanalyses with the published results and report whether our results are consistent with the claims made in the article.

We document all of our reproduction and robustness checks efforts in Quarto documents for full transparency, as well as to allow other researchers to examine our results and conduct their own robustness checks.

2.1 Original and Reproduction Study Features

We used the same functions and R packages as the original authors to ensure maximal comparability. However, the original study does not report the versions of R or of the R packages used, which prevents a direct version-to-version comparison. To facilitate future reproducibility efforts, we provide a full list of the R version, package versions, and system information in our supplementary materials. No substantive differences in the analysis procedures were introduced; any minor adjustments or reconstructions of missing scripts are described above.

2.2 Involvement of Original Authors

We did not contact the authors as part of this reproduction. The corresponding author of the original study was sent an earlier version of our reproduction report in March 2026 by the Institute for Replication, giving the authors the opportunity to make any comments on the report prior to its publication. At the time of submission, we had not received a response.

3 Results

3.1 Preregistrations

3.1.1 Discrepancies Between Pre-analysis Plan and Article

The first challenge in comparing the preregistered analyses with the analyses reported in the article consisted in matching the preregistrations with the relevant study and results. We begin this section by reporting on our understanding of these matches before proceeding with a study-by-study comparison of pre-analysis plans and reported analyses.

The article provides links to the following preregistrations for three experimental studies:

- Study 1: https://osf.io/bfhrn (06 June 2019)
- Study 2: https://osf.io/t928m (10 February 2020)
- Study 3 (passives): https://osf.io/drb6f/overview (11 October 2021)

Study 4 (Chinese Study) was not preregistered and is reported as such in the paper (Brough et al. 2024: 1712). In addition, a secondary analysis of the memory naming task (that was part of Studies 1–3) was preregistered a few years later: https://aspredicted.org/jf9cn.pdf (17 November 2023).

The preregistrations for the Rating Studies are distributed across three documents:

- **Agency Study, transitive condition**: "Does mention order influence perceptions of agency in conjoined phrases?" registered on 10 December 2020 at https://osf.io/zp5d4/overview.
- **Agency Study, conjoined condition**: "Does mention order influence perceptions of agency?" registered on 23 November 2020 at https://osf.io/rh9sd/overview.
- **Agency Study, passive condition and Empathy Study**: "Intuitions analysis for Like Me study" (Study 1: judgements of agency in passive constructions; Study 2: judgements of empathy in conjoined, transitive, and passive sentences), registered on 21 November 2023 at https://aspredicted.org/7698v.pdf.

Although every preregistration document was available to us, the distribution of the preregistered studies and conditions did not correspond to the organisation of the Rating Studies in the paper, making it challenging for us to reconstruct the mapping between the preregistered analyses and the reported results. As far as we can tell, the second hypothesis mentioned in the preregistration for the transitive condition ("Based off the previous experiment, we also predict that participants will be more likely to rate the first-mentioned name as agentive if it is a man's name rather an a woman's name.") was not tested or at least not reported. We also noted that the interaction between the gender of the first-mentioned name and participant gender was not tested in the available analysis scripts, nor is this interaction effect reported in the article.

3.1.1.1 Pre-analysis Plan of Study 1

For Study 1, we found that the analysis code matches the descriptions of the analysis plan in the preregistration (https://osf.io/bfhrn). The authors created two subsets for each of the gender and race trials, computing four separate generalised linear mixed-effect models (GLMMs) with `syntax` as a fixed predictor. Although the exact syntax of these models is not provided in the preregistration, the code for the GLMMs reflects the description:

$$\text{OverallLikeMe} \sim 1 + \text{syntax} + (1+\text{syntax}|\text{subject}) + (1+\text{syntax}|\text{event})$$

This model was applied to the overall data, separately to the gender and the race trials, and to all possible gender and race subsets (white man, white woman, Black man, Black woman). The preregistered inference criteria reads: "We will use the standard $p < .05$ criteria for determining if the regressions suggest that the results are significantly different from those expected if the null hypothesis were correct." Given that a total of seven GLMMs were conducted on the same

dataset or subsets of the data, we were surprised that no mention was made of any p-value corrections.

3.1.1.2 Pre-analysis Plan of Study 2

For Study 2, the analysis code differs from the preregistration (https://osf.io/t928m) in several ways. The provided code replicates the same models from Study 1 for the new data collected with improved stimuli in Study 2. A generalised linear mixed-effects model was fitted to the overall data, separately to the race and the gender trials, and to the gender and race subsets. The Methods section for Study 2 (Brough et al. 2024: 1712) explains that the same regression models as in Study 1 were used. The preregistration, however, describes a more complex modelling strategy. Several preregistered analyses, including valence effects, questionnaire-based regressions, interaction tests, and secondary control analyses, were not implemented in the available code. For the primary analysis, the authors preregistered an iterative approach to model building, looking at predictors of using the dominant group (man or white person) first. They start by including participant race and gender (a), then trial type (b), and finally syntax (c) as predictors of first mention of the dominant group:

> Man First/White First ~ Participant Gender * Participant Race + (1 | Subject) + (1 + Participant Gender + Participant Race | Event)
>
> Man First/White First ~ Participant Gender * Participant Race * [Gender Trial Type or Race Trial Type] + (1 + [Gender Trial Type or Race Trial Type] | Subject) + (1 + Participant Gender + Participant Race | Event)
>
> Man First/White First ~ Participant Gender * Participant Race * [Gender Trial Type or Race Trial Type] * Syntax + (1 + [Gender Trial Type or Race Trial Type] + Syntax | Subject) + (1 + Participant Gender + Participant Race | Event)

None of these models, however, are provided in the analysis code and they are not mentioned in the paper.

Concerning the 'like me' analysis, the preregistered analysis plan specified mixed-effects logistic regression models including participant gender, participant race, trial type, syntax, and their interactions in a similar iterative approach:

> Like Me Gender/Race ~ Participant Gender * Participant Race + (1 | Subject) + (1 + Participant Gender + Participant Race | Event)
>
> Like Me Gender/Race ~ Participant Gender * Participant Race * [Gender Trial Type or Race Trial Type] + (1 + [Gender Trial Type or Race Trial Type] | Subject) + (1 + Participant Gender + Participant Race | Event)
>
> Like Me Gender/Race ~ Participant Gender * Participant Race * [Gender Trial Type or Race Trial Type] * Syntax + (1 + [Gender Trial Type or Race Trial Type] + Syntax | Subject) + (1 + Participant Gender + Participant Race | Event)

However, the analyses in the provided code only include syntax as the sole fixed effect, with participant demographics and trial type addressed only via data subsetting rather than factorial modeling:

> OverallLikeMe ~ 1 + Syntax + (1 + Syntax | Subject) + (1 + Syntax | Event)

For both of the described analyses, the authors preregistered additional models with valence condition as a fixed effect, along participant gender, participant race, trial type and syntax, as predictors of the man first, white first, Like Me Gender and Like Me Race biases. These models are neither provided in the analysis code, nor mentioned in the paper.

For the final analysis of Study 2, Brough et al. (2024) preregistered regression models to analyse questionnaire data that contains participants' attitude towards homosexual and interracial relations, using them as predictors of demonstrating Like Me Race bias. As far as we could see, these models are not in the provided code either. As with Study 1, neither the preregistration nor the code or the paper seem to make any provisions to correct for multiple testing.

3.1.1.3 Pre-analysis Plan of Study 3

For Study 3, the preregistration (https://osf.io/drb6f/) differs considerably from the reported analysis. The authors preregistered valence as an independent variable with the levels intimate, casual and negative. As far as we can tell, this variable is not mentioned in the article. The authors also preregistered "exploratory models for each of the valence conditions" which we were unable to locate in the code or paper.

The primary hypotheses concern whether participants first mention the figure belonging to what the original authors describe in the preregistration as the "more socially dominant group", that is, the man or the white figure. For the 'man first' analysis, male participants were expected to mention the man before the woman more often than chance, whereas female participants were expected to be significantly less likely to do so. Following the same logic, the 'white first' analysis predicted that white participants would mention the white figure first more often than chance, whereas Black participants would be less likely to do so. Two regression models were preregistered to test these hypotheses:

> Man First ~ Participant Gender * Syntax + (1 + Syntax | Subject) + (1 + Participant Gender | Event)
>
> White First ~ Participant Race * Syntax + (1 + Syntax | Subject) + (1 + Participant Race | Event)

Similar to Study 2, these models are not reported in the article, nor are they, as far as we could tell, implemented in the analysis code.

Regarding the 'like me' effect, the original authors preregistered their hypothesis that participants would most likely first mention the figure that matches their gender (which, according to the authors, would indicate a 'like me' gender bias) or their race (demonstrating a 'like me'

race bias). To test this, they specified two mixed-effects regression models with participant gender or participant race, respectively, as a fixed effect. These were designed to test whether the 'like me' effect depends on participant gender/race (that is, whether one gender/race shows a stronger 'like me' effect than the other) and, additionally, whether that effect is moderated by `syntax` (interaction of participant gender/race and syntax).

> Like Me Gender ~ Participant Gender * Syntax + (1 + Syntax | Subject) + (1 + Participant Gender | Event)
>
> Like Me Race ~ Participant Race * Syntax + (1 + Syntax | Subject) + (1 + Participant Race | Event)

However, these models were not reported in the article. Similar to Study 2, the analysis in the provided code includes `syntax` as the only fixed effect:

> OverallLikeMe ~ 1 + syntax + (1+syntax|subject) + (1+syntax|event)

Instead, the authors performed and reported subgroup analyses for the groups of Black women, Black men, white women, and white men.

The preregistration also includes preregistered analyses of questionnaire data, which we were not able to locate in the analysis code and which are not discussed in the published article.

3.1.1.4 Pre-analysis Plan of the Rating Studies

Our recreation of the empathy analysis data on the basis of the preprocessing script (`Data Preprocessing/EmpathyStudy/process_data.R`) and the raw data (`Data Preprocessing/EmpathyStudy/full`) confirmed that the dataset was derived from the original 128 participants, with 19 participants who failed 2 or more catch trials excluded. This differs from the preregistration (https://aspredicted.org/7698v.pdf), which specified excluding participants who answered more than 2 catch trials incorrectly. No additional exclusion criteria appear to have been applied. This potentially differs from the preregistration, which stated that trials with response times more than 3 SD from the mean, participants whose self-reported gender did not match Prolific records, and participants who had reported not to have lived in the UK for the last five years would be excluded. While these additional exclusion criteria might not have been relevant or may have been applied earlier on in the process, we note that the preprocessing code of the Empathy Rating Study does not check for them.

By contrast, all of these preregistered exclusion criteria are applied in the processing scripts for the Rating Studies featuring transitive and conjoint constructions, except that, as with the Empathy Study, the exclusion criteria for failed catch trials implemented in the preprocessing scripts is also > 2 rather than ≥ 2 failed catch trials as specified in the corresponding preregistrations (https://osf.io/rh9sd/ and https://osf.io/zp5d4/). The processing script for the passive Rating Study does not include checks for any potential exclusion based on demographics.

The main claims of the paper are based on comparisons of models made with aggregated data from the various Rating Studies. As far as we can tell, these were not preregistered.

3.1.1.5 Pre-analysis Plan of the Omnibus Studies

For the Multi-Study analyses, the preregistration does not appear to match the actual methodology for the analyses conducted across the first four Studies. The preregistration plan on AsPredicted (https://aspredicted.org/jf9cn.pdf) describes a different experiment that is part of the paper, namely the study on participants' reaction times, where the authors measure whether the participants' response speed is predicted by whether the character matches the participant in either race or gender. The actual preregistration contains the following analyses:

> For each accurate 8AFC response with RT $<$ 10s, we predict logged RT in a mixed effects model by
>
> 1) Whether the character matches the participant's race _and_ gender (coded 1 in a dummy variable) or not (coded 0), i.e.,
>
> lmer(log(rt) ~ OverallLikeMe + (1+OverallLikeMe|id)
>
> 2) Whether the character matches the participant's race i.e.,
>
> lmer(log(rt) ~ like_me_race + (1+like_me_race|id)
>
> 3) Whether the character matches the participant's gender i.e.,
>
> lmer(log(rt) ~ like_me_gender + (1+like_me_gender|id)
>
> 3) Whether the character matches the participant's gender and race exactly (coded 1), matches only one feature (coded 0), or matches no features (coded -1)
>
> lmer(log(rt) ~ like_me_graded + (1+like_me_gender|id)

Although these analyses are conducted in a different section of the paper, the main omnibus analyses reported in the paper (see Brough et al. 2024: 1714 for model formula), which form the basis for the most important claims of the paper, appear not to have been preregistered.

3.2 Availability of Materials

The original data and code were made available on the OSF (https://osf.io/87vew). The repository includes a large number of raw and analysis data files, as well as data processing and analysis scripts in R. Given that the original publication comprises fives studies, we first spent a considerable amount of time matching studies to subdirectories and files. The absence of a README file made navigating the multitude of replication packages challenging.

3.3 Computational Reproducibility

In the preprocessing scripts for Studies 1–3 (`Process_Study1_Data.R`, `Process_Study2_Data.R`, `Process_Study3_Data.R`) the authors mention that, due to a technical issue involving an unbacked-up computer, the authors no longer have access to all the initial raw data . However, they provide full access to data files which have apparently only been manually corrected for minor spelling errors (e.g. 'Rth' instead of 'Ruth'). We therefore classified these files as "raw data provided" in the summary of our computational reproduction (Table 1).

Table 1: Summary of our computational reproduction

	Fully	**Partial**	**No**
Raw data provided	x		
Cleaning code provided	x		
Analysis data provided	x		
Analysis code provided	x		
Reproducible from raw data		x	
Reproducible from analysis data		x	

We were also able to successfully reproduce all of the plots presented in the paper (see `OSF_Analyses_reproduction.qmd`).

3.3.1 Studies 1–3

We were able to computationally reproduce the results of Studies 1–3 with the code provided by the authors. The only noticeable difference was a slight change in the numbers of the 95% bootstrapped confidence intervals with a maximal difference of ± 0.2, which we believe is due to the way random seeds were set in the R scripts. The function `set.seed()`, by default, in R only applies to the next operation that involves randomisation. The authors only set a seed at the top of the script, which only controlled the randomisation for the first function.

We successfully recreated the processed dataset necessary for the analysis of the memory naming task (`studies1_2_3_memory.csv`) based on the raw data using the script `overall_memory_data_processing.R`; however, we noted that this script – which is over 1000 lines long – appears to repeat many of the data processing steps completed in earlier data processing scripts. This is particularly true of the removal of trials and participants according to the exclusion criteria; the latter of which is achieved by filtering out hard-coded participant IDs, which were presumably identified via the visual analysis of the tables produced in the script. It also includes some interactive elements (where the user needs to change a variable multiple times to check various subsets of the data) and many repeated code chunks within the script, too, making it difficult to check for consistency. In spite of the presence of comments

in the code, the logic of the various data wrangling steps is not obvious. As delivered, this preprocessing script is not reproducible.

The main result of the memory naming experiment could be reproduced (= 0.02 (0.005), t(642) = 4.5, p < 0.001). The mean reaction times reported on pp. 1709–1710 could all be reproduced. The bootstrapped confidence intervals differed slightly, but this is not surprising as no random seed was set in the `overall_memory_analysis.R` script.

3.3.2 Study 4

Concerning Study 4, we were unable to fully reproduce two processed data files (`study5_chinese.csv` and `data5_chinese.csv`) from the raw data using the corresponding preprocessing script (`Data Preprocessing/Chinese Study/Process_Chinese_Data.R`). The script returned an error because the column `TrialType` does not exist. We noted that the raw data does include a column entitled `trial_type` which, however, only contains a single unique value: "survey_text". By contrast, the `Trial_Type` variable contained in the processed data file uploaded by the authors, `data5_chinese.csv`, has two levels, differentiating between "RaceTrial" and "GenderTrial". We were unable to reproduce this variable on the basis of the raw data, but all other variables were successfully reproduced. For Study 4, we therefore had to use the `study5_chinese.csv` and `data5_chinese.csv` processed data files provided by the authors to reproduce the statistical analyses based on these data.

Moreover, we were unable to reproduce the median time taken to complete inputting the "critical sentence", which according to the paper is 9.5 seconds in the English condition and 8.3 seconds in the Chinese condition (Brough et al. 2024: 1713). It was not clear to us what exactly was meant by "critical sentence" and how these could be identified in the datasets. We also observed a minor difference in the final sample size for women in the English condition for Study 4. The paper refers to 28 women (2024: 1713), but after excluding those with slow RTs, the script `Process_Chinese_Data.R` outputs a table with `condition`, `ppt_gender`, `list` and `n` in which only 27 English women are listed.

We were able to reproduce the results of the 'like me' effect reported as part of Study 4. However, we were unable to locate the code necessary to reproduce some of the values reported in the paper and therefore had to write the code ourselves to check these numbers. This was the case for size of the overall 'like me' effect, reported as $\beta = 0.18$ (0.04), z = 5.0, $p < 0.001$, which we successfully reproduced, as well as the claim that "the size of the effect did not significantly differ depending on the language used" (Brough et al. 2024: 1710), for which we reproduced the same statistics as in the original study apart from one value: we obtained a negative number for the z-value (z = -1.299) whereas, in the paper, a positive z-value ($z = 1.3$) is reported (Brough et al. 2024: 1710). This sign error is likely a typographical error and does not change the claims made in the paper. As with Studies 1-3, we also observed minor differences in the values of the 95% bootstrapped confidence intervals.

3.3.3 Rating studies

For the Empathy Rating Study, the preprocessing script (`Data Preprocessing/EmpathyStudy/process_data.R`) filtered participants, but did not create the processed dataset used in subsequent analyses. Consequently, it was originally unclear to us whether the data used in the analysis script (`Data Analysis/empathy_study_data.csv`) matches the output of the preprocessing steps. However, we were eventually able to successfully recreate the dataframe used by the original authors, based on an modified version of their preprocessing script, the information provided in the preregistration (https://aspredicted.org/7698v.pdf), and the raw data provided by the authors (`Data Preprocessing/EmpathyStudy/full_empathy_data.csv`).

The Agency Rating Study is based on two separate preregistered studies and data collections (one for transitive and one for conjoint constructions), both carried out on Prolific. We were able to successfully reproduce the combined processed dataset from the raw Prolific data using the preprocessing script provided (`Data Preprossessing/Rating Study/process_rating_data.R`).

On p. 1709, Brough et al. (2024) write:

> As Fig. 2a shows, the sizes of the like me effects varied only minimally across the grammatical constructions that we had tested ($\chi^2(2) = 0.65$, P = 0.721). Rated agency and rated identification, by contrast, strongly varied based upon the grammatical constructions (both $\chi^2(2) > 200$, P < 0.001) and clearly dissociated from one another.

We were unable to reproduce the first chi-square value. Our understanding of the code is that the "Test of grammar" section (from line 1022 of the authors' `OSF_Analyses.R` script) best corresponds to this analysis. This test also returns a non-significant p-value (0.4886), but a chi-square value of 2.4272. We were able to reproduce the second chi-square value in the quoted paragraph above as 201.99 for the rated agency (transitive and conjoint sentences) and 820.96 for the rated identification (referred to in the code/data as the Empathy Rating Study). Both correspond to p-values that are, indeed, < 0.001.

3.3.4 Multi-Study Analyses

For the Multi-Study analysis, which combines the data across Studies 1 through 3, the Methods section (Brough et al. 2024: 1712ff.) describes what is being investigated in that part of the paper:

> We analysed the effect of syntax across the first three studies using a regression of the form LikeMe ~ 1 + Grammatical Frame + (1|subject) + (1|event).
>
> [...]

> The main omnibus analyses used mixed-effects regressions of the form LikeMe ~ 1 + Participant Gender × Participant Race + (1|subject) + (1 + Participant Gender × Participant Race|event). Predictors were contrast coded, and separate models were fit to the race and gender trials. The analysis of trial type (using data from all four production studies) had the form LikeMe ~ 1 + Trial Type [Gender versus Race-focused] + (1 + Trial Type|subject) + (1 + Trial Type|event)" (2024: 1713f.).

These regression models can be found in the code provided by the authors in `OSF_Analyses.r`. They match the descriptions in the paper and we were able to successfully reproduce the results of these analyses.

3.4 Robustness Checks

We conducted a total of four robustness checks. In the following, we briefly report on the outcomes of the first three (for further details, see our full reproduction report in Klünter et al. (2026) and supplementary materials), before focusing on our final robustness check concerning the memory naming task on which the main claims of the article are based.

3.4.1 Reanalysis of the Study 2 with the Preregistered Models

For Study 2, we fitted the following models to the gender trial data and the race trial subsets separately, as preregistered (see Section 3.1):

> Like Me Gender ~ Participant Gender * Participant Race + (1 | Subject) + (1 + Participant Gender + Participant Race | Event)

> Like Me Race ~ Participant Gender * Participant Race + (1 | Subject) + (1 + Participant Gender + Participant Race | Event)

In both cases, the maximal random-effects structure specified in the preregistration failed to converge. In line with the preregistration, which specified that random effects accounting for the least variance and interactions preventing convergence would be iteratively removed, we simplified the random-effects structure by first removing the random slopes for participant gender and participant race at the event level, since the event-level slopes have near-zero variances:

> Like Me Gender/Race ~ Participant Gender * Participant Race + (1 | Subject) + (1 | Event)

Despite these adjustments and the use of the "bobyqa" optimizer, the function returned singularity warnings, indicating that the remaining random-effects structure was still unsupported by the data. Only after removing all random effects did the models converge with no singularity warning. However, we deem this simplification as highly problematic, as it eliminates the

Table 2: Study 2 Results of the preregistered 'Like me' model for race trials (convergence-adjusted)

Characteristic	OR	95% CI	p-value
(Intercept)	1.21	1.15, 1.27	<0.001
ppt_genderwoman	0.99	0.94, 1.04	0.7
ppt_racewhite	1.00	0.95, 1.05	>0.9
ppt_genderwoman * ppt_racewhite	0.98	0.93, 1.03	0.4

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

ability to account for individual variance within participants and systematic differences across verbs. Nonetheless, we interpret the converged models as the closest we could reach to the preregistered analyses. To facilitate the interpretation of the fixed effects and to allow comparison with the reported descriptive results, we computed estimated marginal means using the emmeans package. The model output for the race 'like me' effect Table 2 indicates that this model explains virtually no variance in the data (Tjur's $R^2 = 0.000$). None of the fixed effects, i.e. participant gender, participant race, or their interaction, significantly predict the likelihood of showing a 'like me' effect on race trials. This suggests that, under the preregistered modelling framework, there is no detectable evidence for systematic variation in 'like me' biases on race trials across participant groups.

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3.4.2 Reanalysis of the Study 3 with the Preregistered Models

For Study 3 we performed a robustness check by computing the preregistered models using the original data. For the gender trials, the preregistered model yielded almost identical results to the model reported in the article. In both the preregistered and the reported model, participants chose the same-gender character first in 55.9% of trials. The intercept was significant, confirming an overall 'like me' effect in gender trials (i.e. both men and women were, on average, biased toward their own gender). Fixed effects (participant gender, syntax and their interaction) had no significant effects, in line with Brough et al. (2024: 1709)'s statement that there was no significant difference between men and women concerning their 'like me' effect and no difference between active and passive sentences.

The analogous preregistered race model, fit to race trials, showed no clear evidence for a 'like me' effect. Again, the values computed by the preregistered model match closely the ones reported by the authors (Table 4). Participant race and syntax were not significant, neither was their interaction, indicating no significant difference between white and Black participants and no difference between active and passive in race trials.

The maximal preregistered models only converged when using the "bobyqa" optimizer and produced boundary (singular) fit warnings, indicating that the random effects explain little to no variance. This may explain why the authors decided not to report the preregistered models. In sum, implementing the preregistered model as a robustness check of Study 3 had no effect on the magnitude or the statistical significance of the main point estimates.

3.4.3 Reanalysis of the Study 4 excluding two potentially problematic stimuli

In their discussion of Study 4, Brough et al. (2024: 1713) indicate – presumably in response to a comment from a reviewer[1] – that the observed 'like me' gender effects may be smaller in Chinese due to the inclusion of two potentially gender-neutral Chinese names in the experiment (霍普 [Hope] and 贝斯 [Beth]). We therefore decided to rerun the authors' original analysis for Study 4 with a subset of the data that excludes these two neutral names. We also noticed that the main claim made on the basis of these data relies on a GLMM in which the 'like me' effect is modelled as a numeric predictor ranging from -1 to 0. However, this variable only has three possible values that, somewhat counterintuitively, correspond to the following:

- -1 = highest me-likeness (i.e. both race and gender match participant demographic)
- 0 (only race or gender matches participant demographic)
- 1 = lowest me-likeness (neither race nor gender match participant demographic

We believe that this variable would be better modelled as an ordinal categorical variable, hence we also reran the reported model with this predictor encoded as an ordered factor. Our results confirm that, when excluding neutral names, the 'like me' effect are larger than those reported in the paper .

3.4.4 Reanalysis of Study 5

We noticed that the main claim made on the basis of these data relies on a GLMM in which the 'like me' effect is modelled as a numeric predictor ranging from -1 to 0. However, this variable only has three possible values that, somewhat counterintuitively, correspond to the following:

- -1 = highest me-likeness (i.e. both race and gender match participant demographic)
- 0 (only race or gender matches participant demographic)
- 1 = lowest me-likeness (neither race nor gender match participant demographic

[1]The peer review files are available on https://media.springernature.com/original/springer-static/esm/art%3A10.1038%2Fs41562-024-01943-3/MediaObjects/41562_2024_1943_MOESM2_ESM.pdf (last access 24 October 2026).

Table 3: Summary of our robustness reproduction model with ordered factor predictor variable

Characteristic	**Beta**	**95% CI**
(Intercept)	7.9	7.9, 7.9
ordered(LikeMeFeatureNumeric)		
ordered(LikeMeFeatureNumeric).L	0.03	0.02, 0.04
ordered(LikeMeFeatureNumeric).Q	-0.01	-0.02, 0.00
id SD (Intercept)	0.27	
id SD.LikeMeFeatureNumeric	0.06	
id cor (Intercept).LikeMeFeatureNumeric	0.11	
Residual SD	0.43	

Abbreviation: CI = Confidence Interval

We believe that this variable would be better modelled as an ordinal categorical variable, hence we reran the reported model with the independent variable modelled as an ordered factor.

The coefficient of interest turns out to be robust to this manipulation. It is of a similar size to the reported coefficient and also statistically significant (see Table 3). Note that, in both the model reported in the paper and this additional model, the `LikeMeFeature` predictor accounts for just 0.1% of the variance in participants' recall times. We were surprised that this important statistic was not reported in the paper.

In the original analysis of the memory naming task, the authors removed trials for which participants needed longer than 10 s to name the images (as preregistered). Below are the descriptive statistics of the response time variable for the trials that were included in this analysis (i.e. first guesses only, correct responses only, and excluding participants with strong biases). It turns out that 10,000 ms is much longer than most people needed to recall the names (Q1 = 1891 ms, median = 2627 ms, mean = 3663.3855765 ms, Q3 = 4039.675 ms).

At the same time, we noted that the `overall_memory_analysis.R script` produces a plot which illustrates the difference in response time across the four conditions. This plot, which is not included in the final version of the paper, shows that the differences in mean response times are very small, that the confidence intervals overlap, and that there is considerable variation across the entire dataset (see Figure 1). In fact, not all of this variation is visible as the plot cuts the x-axis off at 4,500 ms, thereby removing 366 data points.

When we visualise all the data points entered into the reported model as boxplots, a rather different picture emerges (see Figure 2). The pattern in median response time does not match the reported effect.

Bearing in mind the outliers visible in Fig. 2, we reran the analysis with a more conservative cut-off point of 6 s than that of 10 s preregistered and applied by the authors (corresponding to 91% of the trials analysed by authors) to test the robustness of the reported effect. The

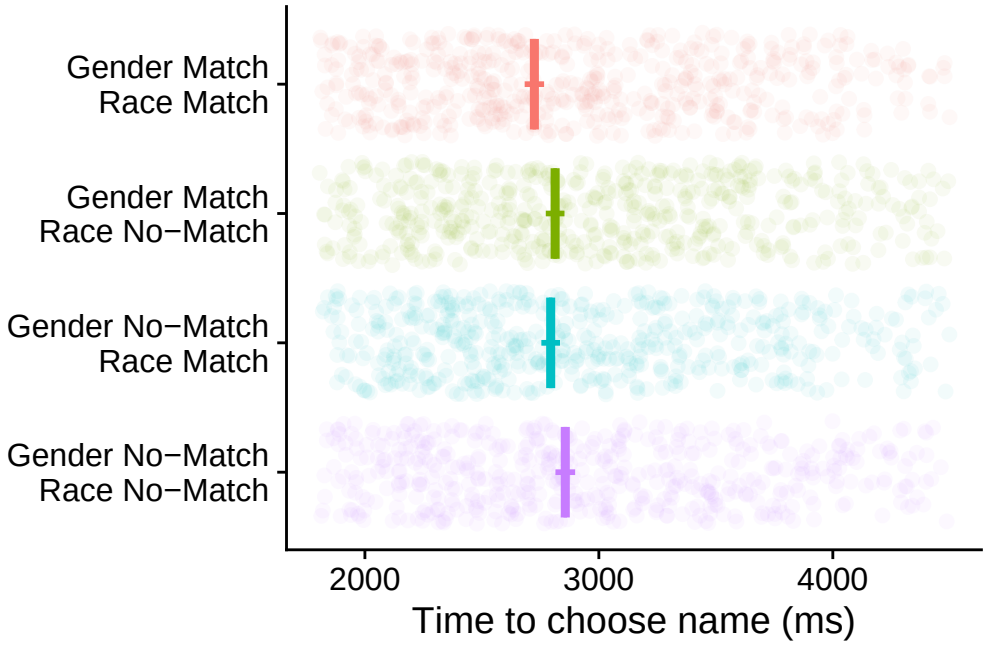


Figure 1: Visualisation showing response time across the four conditions (as coded in original analysis script

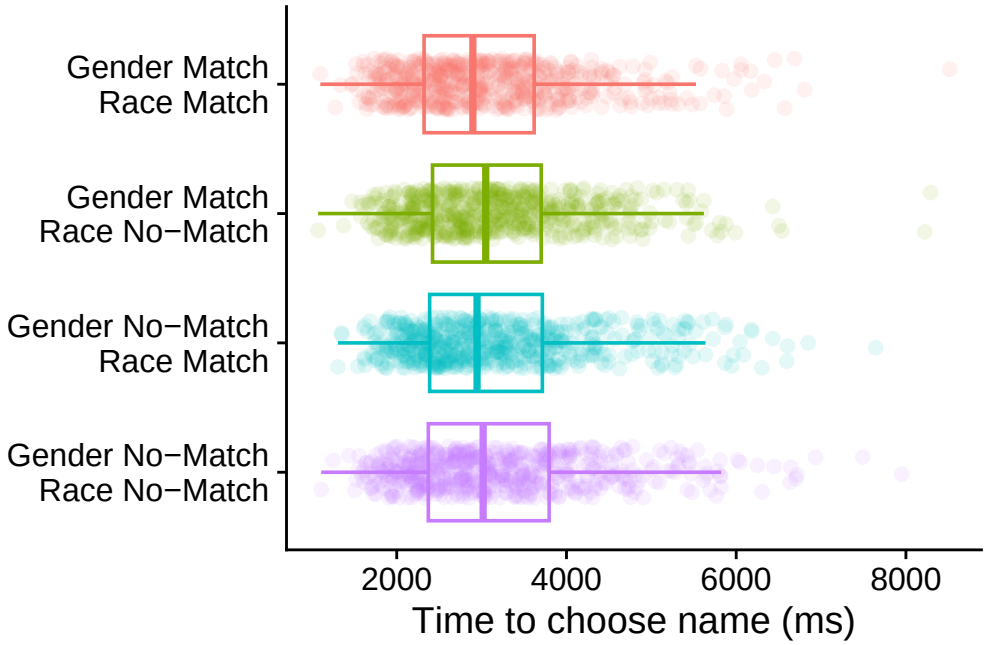


Figure 2: Our own visualisation showing response time across the four conditions

differences in both observed and predicted mean reaction times across the three conditions are smaller than those reported, but remain nonetheless significant. We also reran the model on this subset of the data. The results remain broadly the same (see Table 4).

Table 4: Results of robustness check on naming memory task analysis

Claim in the original study (as reported on pp. 1709-1710)	**Computational reproduction**	**Robustness check**
Response times significantly increased as demographic match decreased	$\beta = 0.020656$ (0.004641) $t(642) = 4.451$ $p = 0.0000101$	$\beta = 0.017077$ (0.003973) $t(613) = 4.298$ $p = 0.00002$
Participants were fastest to select names when they matched the figures on both gender and race	2,724 ms [2,642–2,801]	2,548 ms [2,488–2,612]
When participants matched the figures in gender but differed in race (and, thus, the forms of names were perfectly counterbalanced), then responses were about 90 ms slower	2,813-2,724 = 89 ms	2,639-2,548 = 91 ms
When participants matched in race but differed in gender, then responses were about 70 ms slower	2,794-2,724 = 70 ms	2,605-2,548 = 57 ms
Finally, when participants differed from the figures in both gender and race, they were an additional 50 ms slower to name them	2,856-2,794 = 62 ms	2,662-2,605 = 57 ms
Model performance metrics		
N	22482	12311
Marginal R^2 / Conditional R^2	0.001 / 0.287	0.001 / 0.282

3.5 Dataset and Materials link

3.6 Object/file name

3.7 FAIR data/Codebook

- ☐ Data deposited in certified repository (e.g., OSF, Zenodo)
- ☐ Dataset has persistent identifier (DOI)
- ☐ Code shared with open license (e.g., MIT, GPL)
- ☐ Codebook includes:

- ☐ Variable names Description
- ☐ Valid ranges
- ☐ Missing codes

☐ Metadata uses standard schema (e.g., Datacite, CodeMeta)

☐ File formats are non-proprietary (e.g., .csv, .txt, .R)

4 Evaluation

Overall, the computational reproduction of Brough et al. (2024) was largely successful. The authors shared raw data, data preprocessing code, processed data, and analysis code on the OSF, and we were able to reproduce nearly all of the reported statistics and all of the plots. Where discrepancies arose, they were small and did not change any of the article's claims. In particular, we reproduced the central result on which the main claims rest, the effect of demographic match on response times in the memory naming task, which also held up in our robustness checks (see Table 3 and Table 4). However, we identified three potentially critical issues:

1. As far as we can tell, no correction for multiple testing was applied, neither in the preregistrations nor in the analyses, even though many models were fitted to the same data and subsets of the data.
2. For Studies 2 and 3, the preregistered analyses differ considerably from the reported ones, and the main omnibus analyses that underpin the paper's most important claims do not appear to have been preregistered at all. This is not in and itself problematic, but is not explicitly stated in the paper (unlike for Study 4).
3. The fact that the reported effects that support the paper's main claims are very small is not made sufficiently clear in the article.

The materials provided by the authors allowed us to reproduce the main results of Studies 1–4 and the Rating Studies, with only minor discrepancies that do not affect the article's claims. These include slightly different bootstrapped confidence intervals (likely due to how random seeds were set), a sign difference in one z-value, a chi-square statistic in the Rating Studies that we could not reproduce (but which was non-significant in both cases), and a few values and variables that we could not recover from the raw data.

The main obstacles to computational reproducibility were organisational rather than analytical. The repository has no README, some of the preprocessing scripts are repetitive, partly interactive and/or rely on hard-coded participant IDs. Part of the raw data for Studies 1–3 is no longer available, and software versions are not reported. Reproducing the results therefore required considerable effort in matching studies, scripts, and preregistrations and not all preprocessing decisions are transparent.

For Studies 2 and 3, the preregistered models were not implemented in the code or mentioned in the article. Combined with the lack of multiple-testing corrections, this makes it difficult for readers to distinguish confirmatory from exploratory analyses and increases the risk that some of the reported significant results may be false positives.

Our robustness checks largely supported the reported findings. When fitted to the original data, the preregistered models for Study 3 returned almost identical estimates to the reported ones. For Study 2, we found no evidence that the 'like me' effect varies with participant gender or race, although our models only converged after we removed all random effects. In the memory naming task, the effect remained significant when we modelled the predictor as an ordered factor and when we applied a stricter response-time cut-off, although the differences between conditions became smaller. However, we show that alternative plotting of the data entered into the model do not clearly reflect the reported pattern in median response times. We conclude that the reported effect is statistically detectable but very small, suggesting the article's framing of it as an important driver of biased language production deserves more cautious interpretation.

We did not collect new data and cannot assess generalisability. We note, however, that the claim that the effect is independent of language rests on only two languages, as Brough et al. (2024: 1711) acknowledge. The failed replication by Halpern (2026), although based on a very small and WEIRD sample, further suggests that the effect may not be as robust across settings as claimed. We suggest that more replication studies are necessary.

4.1 Limitations

We acknowledge that our study has several limitations. First, our reproduction is based solely on the data and code provided by the authors; we did not contact them to clarify ambiguities (e.g. the meaning of "critical sentence" or the origin of `Trial_Type`). Some of the discrepancies we report may therefore have simple explanations that we were unable to uncover.

Second, we were unable to fit the maximal preregistered random-effects structures: the models for Study 2 only converged after all random effects were removed, which we consider problematic, and the Study 3 models produced singular fits. Our robustness checks for these studies are therefore only approximations of the preregistered analyses.

Third, some of our robustness checks involved analytical decisions of our own (e.g. the 6 s cut-off, the ordered-factor coding), which are themselves open to criticism and we did not preregister our robustness checks.

Finally, as noted above, we did not collect new data and thus cannot assess the generalisability of the findings. We also did not examine the preregistered analyses that were missing from the code and article (e.g. concerning valence and the questionnaire data), as the necessary data and models were not available to us.

4.2 Recommendations for Future Studies

Based on our experience, we would like to make the following recommendations.

For authors of similar studies:

- Include a README that clearly maps each study, preregistration, script, and data file to the corresponding results in the paper.
- Report R and package versions (e.g. using `sessionInfo()` or the `renv` package; Ushey and Wickham (2023)) and set random seeds before every stochastic operation.
- Provide linear, annotated preprocessing scripts that run from raw data to analysis data without hard-coded participant IDs or manual, interactive steps, and keep a record of any manual corrections to raw data.
- Report all deviations from preregistrations (including changes to exclusion criteria, models, and variables), and clearly label which analyses were confirmatory and which were exploratory. Preregister the main omnibus analyses and state how multiple comparisons are handled.
- Report effect sizes that are central to the main claims in interpretable units (e.g. milliseconds or percentage points) with a measure of variability around these estimates, especially for very small effects.

For future replications of this research:

- Conduct independent replications with larger and more diverse samples than the one used by Halpern (2026), and with additional languages, to test the claim that the 'like me' effect is independent of language and social group.
- Use stimuli and names that have been previously tested for unambiguous gender neutrality or gender markedness in the target language.
- Preregister analyses together with a smallest effect size of interest.

Future studies could also attempt to test the effect in other tasks and in more naturalistic production settings to examine whether the incrementality-and-accessibility account holds beyond the memory naming paradigm.

4.3 Brief Evaluation

The computational reproducibility of Brough et al. (2024) is high: all main results and plots could be reproduced from the shared materials, with only minor numerical discrepancies and a few unreproducible values that do not affect the article's claims. The main findings were also robust to our re-analyses, although the effect is very small, p-values were seemingly not corrected for multiple testing, and several preregistered analyses were not implemented or reported.

Contribution Statement

AI disclosure

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Conflicts of interest

The author(s) declare no conflict of interest associated with the publication of this manuscript.

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References

Andringa, Sible, and Aline Godfroid. 2020. "Sampling Bias and the Problem of Generalizability in Applied Linguistics." *Annual Review of Applied Linguistics* 40 (March): 134–42. https://doi.org/10.1017/S0267190520000033.

Axt, Jordan, Valentina Palacio Posada, Eliane Roy, and Jeffrey To. 2025. "Revisiting the Policy Implications of Implicit Social Cognition." *Social Issues and Policy Review* 19 (1): e70003. https://doi.org/10.1111/sipr.70003.

Brough, Jessica, Lasana T. Harris, Shi Hui Wu, Holly P. Branigan, and Hugh Rabagliati. 2024. "Cognitive Causes of 'Like Me' Race and Gender Biases in Human Language Production." *Nature Human Behaviour* 8 (9): 1706–15. https://doi.org/10.1038/s41562-024-01943-3.

Halpern, Rebekka Jeanne Elisabeth. 2026. "Exploration of Animacy in the "Like-Me" Effect." Bachelor thesis. Florida State University. https://doi.org/10.33009/FSU_f151ee87-6323-450c-a8df-c6e26ae41025.

Hausladen, Carina I, Manuel Knott, Colin F Camerer, and Pietro Perona. 2025. "Social Perception of Faces in a Vision-Language Model." *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency*, June, 639–59. https://doi.org/10.1145/3715275.3732041.

Henrich, Joseph, Steven J. Heine, and Ara Norenzayan. 2010. "Most People Are Not WEIRD." *Nature* 466 (7302): 29–29. https://doi.org/10.1038/466029a.

Hermann, Erik, Julian De Freitas, and Stefano Puntoni. 2025. "Reducing Prejudice with Counter-Stereotypical AI." *Consumer Psychology Review* 8 (1): 75–86. https://doi.org/10.1002/arcp.1102.

Klünter, Marie, Rose Hörsting, Luca Leppert, Gina Reinhard, and Elen Le Foll. 2026. "A Comment on 'Cognitive Causes of "Like Me" Race and Gender Biases in Human Language Production' (Brough Et Al. 2024)." https://doi.org/tba.

Röseler, Lukas, Lukas Wallrich, Helena Hartmann, Luisa Altegoer, Veronica Boyce, Sarahanne M. Field, Janik Goltermann, et al. 2025. "Handbook for Reproduction and Replication Studies." https://forrt.org/replication-handbook/.

Ushey, Kevin, and Hadley Wickham. 2023. *Renv: Project Environments*. https://CRAN.R-project.org/package=renv.